Lower-Tract GU Malignancies

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Block: Endocrine/Urology//Reproduction

Thread: Clinical Medicine

Conflict of interest: Dr. Shah has no conflicts of interest to disclose.

Further Reading

Respective chapters in:

Campbell-Walsh Urology. In Wein A. J.,editor., Kavoussi L. R.,editor., Partin A. W. & Peters C. (Eds.), (Eleventh edition), 2012.

Glenn's Urologic Surgery. In Graham S. D., Keane T. E. & Glenn, James F. (James Francis),1928-(Eds.), (7th ed ed.). Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins. 2010.

Smith & Tanagho's General Urology. In McAninch, Jack W., Lue, Tom F., Smith, Donald R. . Smith & Tanagho's General Urology. 18th ed. / edited by, Jack W. McAninch, Tom F. Lue. New York: McGraw-Hill Medical; 2013.

Learning Objectives

By exam time, you should be able to do the following:

1. Explain the presented epidemiology, etiology of and risk factors for prostate cancer.
2. Discuss the role of physical exam and PSA in screening for prostate cancer.
3. Describe the process for diagnosis of prostate cancer and explain the diagnostic and prognostic characteristics of Gleason Score for prostate cancer.
4. Describe the treatment options for a patient who presents with localized prostate cancer.
5. Describe the treatment concepts for metastatic castrate sensitive and castrate resistant prostate cancer.
6. Explain the presented epidemiology of and risk factors for bladder cancer.
7. Describe the prognostic and therapeutic differences between muscle-invasive and non-muscle-invasive urothelial carcinoma of the bladder
8. Describe the epidemiology of and risk factors for testis cancer.
9. Describe the presentation, basic components of staging, treatment options, and outcomes for a patient who presents with testis cancer.
10. Identify the most common histologies and tumor markers of testis cancer

Study Questions

1. A 71-year-old patient presents with de novo irritative voiding symptoms. Her PCP checks a urinalysis that demonstrates microhematuria. She is sent to a urologist who performs a cystoscopy that demonstrates a papillary mass in the bladder.
    - What is the most likely diagnosis?
    - What is the next step in diagnosis?
    - What pathological features in a biopsy specimen dictate management?

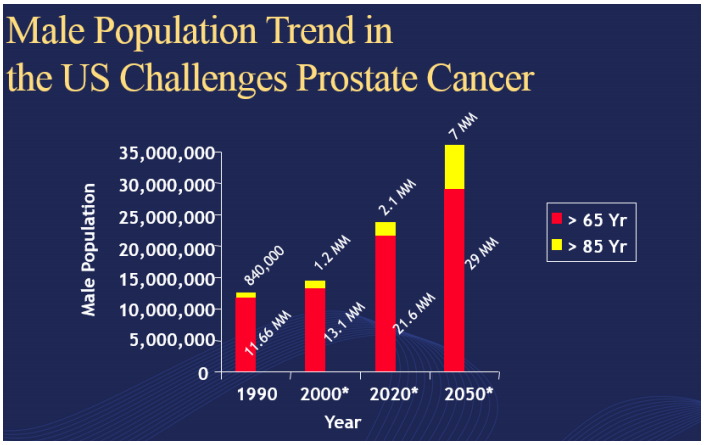
2. A 52-year-old patient is noted on annual screening to have a PSA of 6.2 ng/mL. His DRE demonstrates a normal consistency gland with smooth borders. He is diagnosed with a Gleason Score 4+3 prostate cancer in 2/12 cores biopsied.
    - What is the significance of the PSA level?
    - What is the clinical stage of the tumor?
    - What is the clinical significance of the Gleason Score?
    - What other tests should be ordered on this patient?
    - What treatment options should be offered to this patient?

A. A 32-year-old patient noticed a lump on his testicle during self-examination.
    - What should be your next step in diagnosis?
    - What is the most likely diagnosis?
    - How should this patient be staged?
    - What dictates the treatment options for this patient?

A. Adenocarcinoma of the Prostate

1. Background/Epidemiology

Probability of Developing PC	
Birth to 39	1 in 9,4422
40 to 59	1 in 41
60 to 69	1 in 16
70 and older	1 in 8
Birth to death	1 in 7



~ *80% of new prostate cancer diagnoses are in people >age 65yrs*

2. Risk Factors - Etiology is multifactorial
   a. Androgens
      i. Role of hormones is essentially permissive
      ii. Prepubertal castrated people do not develop prostate cancer
      iii. Cancers respond (for a time) to chemical or surgical castration
   b. Heredity
      i. Increased risk among first degree relatives with prostate cancer
      ii. Highest in Blacks/Scandinavian
      iii. Lowest in Asians
      iv. 2 or 3 first degree relatives increases the risk 5 to 11-fold
      v. Increased risk in relatives with breast cancer

c. Environment

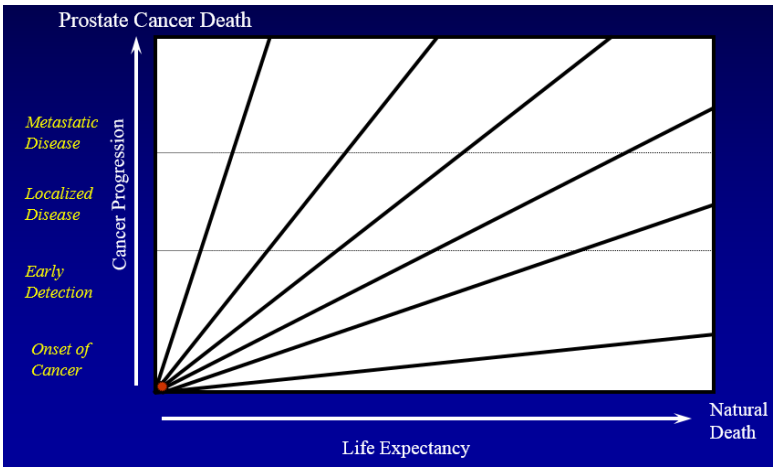
i. Dietary

ii. Asian immigrants have higher risk (not as high as native born Americans)

iii. As Asian diet is "westernized" incidence appears to be rising.

iv. Specific dietary components have not been identified (fats?)

3. Natural History – Heterogenous



4. Screening/Diagnosis

a. DRE

i. Least sensitive way to detect prostate cancer

ii. If DRE is (+) and prostate cancer is detected, it is usually more advanced disease

b. PSA

i. PSA is a single chain serine protease

1. Member of the human tissue kallikrein family

2. PSA is produced for the ejaculate, where it liquefies semen in the seminal coagulum and allows sperm to swim freely.

ii. Elevated PSA

1. Prostate cancer, prostatitis, BPH, urethral/prostatic trauma, sex/ejaculation (rarely), urinary retention. **Aggressive** DRE

iii. Decreased PSA:

1. Antibiotics will lower PSA in 30% if infected

2. Simple repeat will lower in many men

3. Finasteride/Dutasteride lower PSA by 50% (any dose)

a. Staging tool for prostate cancer

i. In general, the higher the PSA, the more advanced the disease (0-10 mg/mL = low risk; 10-20 = int risk; >20 high risk)

b. Marker for treatment efficacy in prostate cancer

i. It is excellent in this situation

ii. After radical prostatectomy, the PSA should be 0 or very low and not increase over time

iii. After radiation, the PSA should decrease to low levels and also not increase over time

c. PSA screening

i. Controversial tool for screening

ii. Risk of overdiagnosis and overtreatment for prostate cancer with significant impact on quality of life

iii. It has been shown that PSA:

a. has decreased the incidence of annual prostate cancer deaths by 44%

b. has decreased the incidence of metastatic disease at presentation by 25%

iv. Currently recommended to be checked annually in all male patients >55 years old (sooner if high-risk for development of prostate cancer)

c. Prostate biopsy

d. PSA and DRE are **not** diagnostic of prostate cancer. They only suggest its presence – if they are concerning, a prostate biopsy is recommended.

e. Performed under ultrasound guidance (transperineal or transrectal approach)

f. 12-core mapping biopsy of prostate is performed. Directed biopsies only performed if lesion detected on imaging.

B. Histology and Grade of prostate cancer

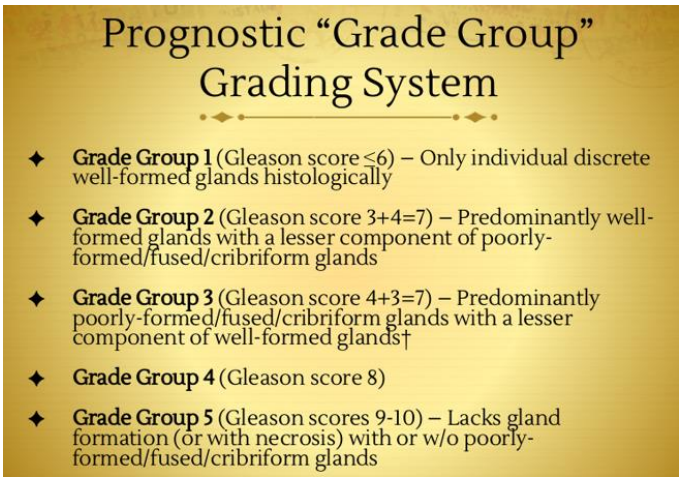
1. Immunohistochemistry for prostate caner

a. Alpha-methylacyl-CoA racemase (AMCAR/p504s) is present in prostate cancer and HGPIN (+ RED)

b. Basal cell layer markers are absent in cancer and present in benign

glands and HGPIN (- BROWN)

2. Gleason Score
    a. Sum of primary and secondary glandular architecture (pattern) of the cancer
3. Patterns 1-5 (Potential Score 2-10)
    a. Aggressiveness increases as Gleason Score increases
    b. Gleason Score ≤ 6 – Least aggressive
    c. Gleason Score 7 – More aggressive
    d. Gleason Score ≥ 8 – Most aggressive



C. Staging
1. Clinical
    a. PSA and DRE
        i. Stage cT1: not clinically detectable, only by core biopsy (T1c-elevated PSA alone)
        ii. Stage cT2: confined to the prostate
        iii. Stage cT3: with extracapsular penetration, prostatic bladder neck or seminal vesicle invasion
        iv. Stage cT4: invades adjacent structures like bladder, rectum, pelvic wall
    b. CT and Bone Scan
        i. Stage cN1: Lymph nodes metastasis

ii. Stage cM1: Distant metastasis

2. Pathologic – **must have tissue**
   a. T Stage
      i. There is no pT1 stage
      ii. pT2: Organ confined
      iii. pT3: Extraprostatic extension
         - pT3a: Periprostatic/ bladder neck invasion
         - pT3b: Seminal vesicle invasion
      iv. pT4: Invasion of rectum, levator muscles and/or pelvic wall
   b. N stage
      i. pNX: Cannot be assessed
      ii. pN0: No regional lymph node metastasis
      iii. pN1: Metastasis in regional lymph node or nodes
   c. M stage
      i. pMx/M0 – cannot be determined/no metastasis
      ii. M1: Distant metastasis
         - pM1a: Nonregional lymph node(s)
         - pM1b: Bone(s)
         - pM1c: Other site(s) with or without bone disease

3. Treatment of Local Prostate Cancer
   a. Driven in part by risk stratification:
      i. Very low risk: T1c, not N1 or M1; Gl ≤ 6; fewer than 3 cores positive with cancer; ≤ 50% cancer in any 1 core
      ii. Low risk: T1c, not N1 or M1; Gl ≤ 6; and PSA <10 ng/ml; but not meeting the very low risk criteria
      iii. Intermediate risk: T1-2, not N1 or M1; Gl ≤ 7; and/or PSA 20 ng/ml; and not meeting very low or low risk criteria
      iv. High risk: T1-4, not N1 or M1; Gl ≥ 8; and/or PSA <100 ng/ml; not meeting other risk groups criteria
   b. Active Surveillance
      i. Only indicated for very low or low risk disease
      ii. Patients must be re-biopsied intermittently (every 6-12 months

initially) to determine for progression of disease into a higher risk category

iii. Patients on Active Surveillance may choose treatment at any time even if they do not progress on re-biopsy

c. Surgery

i. Radical prostatectomy (open, robotic) – the entire prostate, seminal vesicles, and potentially regional lymph nodes are removed

ii. Quality of life implications

- Infertility- permanent (patient will no longer produce an ejaculate upon orgasm)
- Stress urinary incontinence - usually transient
  - Kegels exercises
  - Pelvic floor physical therapy
  - Artificial urinary sphincter
- Impotence – temporary or permanent
  - Nerve preservation – less likely for higher risk disease
  - May respond to PDE5i

d. Radiation (IMRT, brachytherapy) – *less* invasive

i. Entire prostate is radiated (may be extended to include regional lymph nodes)

ii. Similar oncologic outcomes to surgery

iii. Quality of life implications

- Infertility- high (decreased semen production)
- Urinary symptoms – temporary or permanent
  - Usually urinary urgency/frequency
    1. Respond to anticholinergic medications
  - Hematuria – self limiting
- Impotence – temporary or permanent
  - May respond to PDE5i
- Fecal side-effects - temporary or permanent
  - Urgency
  - Bleeding

4. Metastatic disease
    a. Prostate cancer typically metastasizes to lymph nodes and bone
        i. First lymph node landing zone is the obturator and external iliac artery branch
        ii. Bone usually involved is the axial skeleton
    b. Since prostate cancer is a hormonally active cancer (androgens), initial systemic therapy for systemic disease is hormonal: androgen deprivation therapy (ADT) or castration
        i. Surgically, this is accomplished with a bilateral orchiectomy
        ii. Chemically, this is usually accomplished with a GnRH analogue, such as leuprolide
            - GnRH agonists act on the anterior pituitary to create negative feedback and inhibit FSH and LH secretion, which in turn decreases testosterone production in the testis
            - Initially, the agonists create a testosterone surge, which is clinically counteracted by an anti-androgen (androgen receptor antagonist) such as bicalutamide
        iii. Long-term castrate state can have adverse effects on men, including hot flashes, loss of libido/ED, loss of lean muscle mass/increased adipose deposition, metabolic syndrome, anemia/fatigue, osteoporosis/skeletal fractures, arterial stiffness, cardiovascular morbidity, and cognitive decline
            - To counteract some of these adverse effects, men on long-term ADT are encouraged to take supplemental Vitamin D and calcium, engage in weight-bearing exercise, quit smoking, and maintain their general primary care health
    c. Castrate Resistant Prostate Cancer (CRPC)
        i. Eventually, prostate cancer may learn to grow in a castrate state – CRPC.
        ii. CRPC is typically diagnosed by a rise in the PSA and progression of metastatic disease on imaging despite a castrate Testosterone level.
        iii. The etiology of conversion to CRPC is multivariable, and in general portends a poor prognosis
        iv. Treatment of CRPC involves maintenance of ADT along with additional medications that affect androgen production and the androgen receptor (AR)

Bladder Cancer (Urothelial carcinoma)

A. Epidemiology
   1. 70,980 new diagnoses in 2009, 14,330 deaths
   2. 4th most common cancer
   3. 3:1 M:F
   4. Rare <40 years old
   5. Most >65 years old – these patients often have significant medical comorbidities that may affect treatment and prognosis

B. Risk Factors
   1. Behavioral – cigarette smoking
   2. Chemical exposure – arsenic in well water
   3. Chronic infection.
   4. Iatrogenic – cyclophosphamide exposure; pelvic radiation exposure
   5. Occupational – aromatic amines (chemical dyes, pharmaceuticals, petrochemical); manufacturing; service industry

C. Presenting Signs and Symptoms
   1. Hematuria – microscopic and gross
   2. Irritative voiding symptoms - urinary frequency/urgency
   3. Obstructive voiding symptoms – intermittency, decreased stream
   4. Pain associated with locally advanced disease
   5. Constitutional symptoms of systemic disease

D. Histology
   1. Urothelial most common
   2. Non urothelial variants can have poor prognostic implication

E. Grade and Stage
   1. Low grade and high grade
   2. Low grade associated with superficial disease
   3. High grade can be superficial (CIS) or invasive
   4. TNM stage

a. Ta – non invasive, can be either LG or HG
b. CIS – non-invasive, HG
c. T1 – superficially invasive, HG
d. T2 – muscle invasion, HG
e. T3-T4 – locally advanced, HG

F. Treatment
1. Superficial (NMIBC)
a. Transurethral resection of bladder tumor (TURBT) – necessary for diagnosis. Can be therapeutic/curative for superficial disease
b. Intravesical therapy – indicated for recurrent superficial UC (LG or HG). Also indicated for HG NMIBC (CIS or T1).
- BCG
- MMC
- Thiotepa
- Doxorubicin
c. Radical cystectomy with urinary diversion for refractory disease
2. Muscle Invasive (MIBC)
a. Neoadjuvant cisplatin-based chemotherapy given if patient can tolerate
b. Radical cystectomy with urinary diversion

Testis Cancer

A. Epidemiology
1. 2-3 new cases per 100,000 US people with prostates per year
2. Marked variation in incidence among different countries/races
3. 90-95% are germ cell
4. Most common solid tumor in people with prostates ages 15-35

B. Risk Factors
1. Cryptorchidism:
- 7-10% of patients with testicular cancer have a history of cryptorchidism
- Abnormal germ cell morphology
- Elevated temperature

- Interference with normal blood supply
- 5-10% of patients with testicular cancer and a history of cryptorchidism develop cancer in the contralateral testis
- Orchidopexy does not prevent development of cancer – just allows for detection

2. Gonadal Dysgenesis
   - 20-30% develop cancer (gonadoblastoma)
3. Trauma ??
   - prompts evaluation
4. Hormones
   - DES/OCP probably do not increase risk
5. Atrophy (nonspecific vs. mumps orchitis)
   - Speculative

C. Presenting symptoms

1. Painless swelling / mass
2. 30-40% dull / aching sensation
3. 10% present with metastatic symptoms
4. Gynecomastia
   - 5% germ cell
   - 30-50% Sertoli/Leydig
5. 1-2% have bilateral disease at diagnosis
6. More common on the right
7. Infertility

D. Testis Cancer Pathology/Histology

1. Most (~95%) in post-pubertal patients are germ cell tumors
2. Arise from intratubular germ cell neoplasia (common in cryptorchidism and dysgenic gonads)
3. Subclassified as seminomatous and non-seminomatous germ cell tumors.
4. 50% are classic seminoma (histologically identical to ovarian and CNS dysgerminoma).
5. NSGCT include embryonal, yolk sac, choriocarcinoma and teratoma.

6. Mixed germ cell tumors contain more than 1 type usually embryonal, teratoma, yolk sac.

Tumor	Peak age	Morphology	Tumor makers
Seminoma	40-50	Sheets of uniform polygonal cells with clear cytoplsm and stromal lymphocytes	10% HCG
Embryonal	20-30	Poorly diff pleomorphic cell in cords, sheets or papillary formations; can contain yolk sac and chroio	Neg (pure)
Yolk sac	3	Poorly diff endothelium-like, cuboida or columnar	90% AFP
Choriocarcinoma	20-30	Cytotrophoblas and syncytiotrophoblasts without villus formation	100% HCG
Teratoma	All	Tissue from all three germ cell layers with varying degrees of diff	Neg (pure)
Mixed	15-30	Variable	90% HCG and AFP

Stage

7. TNMS

Primary tumor (T)	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Intratubular germ cell neoplasia
T1	Tumor limited to the testis and epididymis or tumor invasion into the tunica albuginea only
T2	Tumor extending through the tunica albuginea with involvement of the tunica vaginalis
T3	Tumor invades the spermatic cord
T4	Tumor invades the scrotum
Regional lymph nodes — clinical (N) or pathologic (pN) staging	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastases to single or multiple lymph nodes, each < 2 cm in size
N2	Metastases to single or multiple lymph nodes, >2 cm but < 5 cm in size
N3	Metastases to lymph node, >5 cm in greatest dimension
Distant metastasis (M)	
MX	Distant metastasis cannot be assessed
M0	No distant metastasis
M1	Distant metastasis
M1a	Nonregional nodal or pulmonary metastasis
M1b	Distant metastasis other than to nonregional lymph nodes and lungs

tumor markers	
S0	Normal level
SX	Unavailable or not performed
S1	Lactate dehydrogenase (LDH) level < 1.5 times normal, human chorionic gonadotropin (HCG) level < 5000 IU/L, alpha-fetoprotein (AFP) level < 1000 ng/mL
S2	LDH 1.5-10 times normal; HCG level, 5000-50,000 IU/L; AFP level, 1000-10,000 ng/mL
S3	LDH >10 times normal; HCG level >50,000 IU/L; AFP level >10,000 ng/mL

E. Evaluation and Workup

1. History and Exam
2. Scrotal U/S
3. CXR +/- Chest CT
4. Abdominal CT
   a. Can identify small nodal deposits <2 cm
   b. MRI and PET scan no advantage over CT
5. Serum markers: AFP, BHCG and LDH
   a. AFP:
      - Expressed by the early embryo (also liver and GI tract)
      - Half-life: 5-7 days
      - Produced by pure embryonal, teratocarcinoma, yolk sac, mixed tumors (NOT pure choriocarcinoma or seminoma)
      - Falsely elevated in liver dysfunction, viral hepatitis and ETOH
      - Presence of AFP in what's thought to be pure seminoma suggests a mixed tumor

b. BHCG:
   - Secretory product of the placenta
   - Alpha unit (LDH,FSH,TSH) and beta unit
   - Half-life: 24-36 hours
   - Produced by syncytiotrophoblastic tissue
   - All choriocarcinomas, 40-60% embryonal, 5-10% seminoma
   - Falsely elevated in hypogonadism and marijuana use

c. LDH:
   - Lactic Acid Dehydrogenase
   - Presents normally in smooth, cardiac and skeletal muscle, liver and brain
   - Most useful in advanced seminoma or tumors where other markers are not elevated
   - Many false positives

d. Elevation after orchiectomy represents metastatic disease

e. Conversely normalization does not rule out metastatic disease

6. Fertility preservation – Sperm Banking for patients with advanced disease in whom additional therapy is planned

F. Treatment

1. Radical orchiectomy
   a. Inguinal approach
      i. Avoids seeding the scrotum and disrupting lymphatics
   b. Wait 5 half-lives before rechecking markers
   c. Biopsy not recommended due to the risk of tumor spread
   d. Necessary for histology identification and possibly curative
2. Seminoma
   a. Stage I – observation versus carboplatin versus RT with RT and/or chemotherapy for relapse
   b. Stage II – RT to the retroperitoneum with/without chemotherapy
   c. Advanced or metastatic disease – cisplatin-based chemotherapy (BEP)
      - 90% will have a complete response

- Residual retroperitoneal masses are usually fibrosis
- RPLND warranted if residual RP mass is >3cm and well circumscribed

3. Nonseminoma
   a. Stage I – observation versus adjuvant chemotherapy versus RPLND dependent on risk stratification
   b. Stage II – chemotherapy versus RPLND dependent on marker positivity. Salvage chemo with RPLND
      - RPLND not indicated for marker (+) disease
   c. Advanced or metastatic disease – cisplatin-based chemotherapy (BEP) with potential for salvage RPLND

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. *This patient most likely has a urothelial carcinoma of the bladder. She will need a transurethral resection of bladder tumor (TURBT) for diagnostic confirmation (as well as treatment). Depending on the grade and stage of the tumor, she can be observed (low grade, superficial), may receive intravesical therapy (high grade, superficial or superficially invasive), or neoadjuvant chemotherapy followed by radical cystectomy (muscle-invasive bladder cancer).*

2. *This patient has a clinical T1c unfavorable intermediate risk prostate cancer. He should be staged with a CT scan of the abdomen and pelvis and a bone scan. In general, the treatment options for localized prostate cancer are active surveillance, primary surgery, or radiation. In select cases, focal therapy (e.g. cryoablation) is also an option. Because of his risk of disease, however, active surveillance is not an option.*

3. *The mass should be confirmed with a scrotal ultrasound. This patient most likely has a germ cell tumor of his testicle. He should undergo a radical orchiectomy. He should be staged with a CT scan of the abdomen/pelvis, either a CXR or CT chest, and tumor markers (AFP, BHCG, LDH). Treatment will be determined based on the histology of the tumor (seminoma versus nonseminoma), and the results of the staging studies (localized, minimal lymph node involvement, gross lymphatic involvement or visceral metastases) and tumor markers (ie if elevated, do they normalize after orchiectomy).*